

Temescal

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Temescal
training

- The temescal tool is a deposition tool for depositing metals onto a sample.

Software

- There are two menus: Auto menu and System Map
 - System Map shows the layout of the system and is used to monitor valves and the operation of the tool.
 - Auto Menu is used for venting and pumping chambers as well as loading and editing recipes.
 - Under the folder `./standards` are all of the standard recipes for materials in the temescal. Copy these files and edit them in your own folder
- it is good practice to immediately start your recipe after you've closed the PC.
- Your recipes should include a pump and vent step before and after deposition.
- Parameters you will be likely to change when you edit a recipe are *final thickness* and *deposition rate*. You may also end up modifying the *soaking*
- Remember to click save when you move the beam on the metal.
 - you move around the beam trying to reduce the power/current.
- in order to open the door, you'll have to press `unlatch` in the system map.
- Always use `down open` as the option when evacuating PC.
- Can't load a recipe in the process of pumping the PC down until the pirani gauge goes green.
- Make sure after you start the execution of your recipe that you press `hold` in order to have control over when the recipe starts.

Hardware

- PC: process chamber. This is the chamber in which you will load a sample.
- SC: source chamber.(you won't have to do anything with this)
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General procedure:

1. vent PC *and Press start*
2. Click `unlatch` to open PC
3. Tape your sample onto a temescal carrier wafer and place upside down on wafer holder in PC.
4. close chamber

5. Pump PC in down open position
6. Once the pirani gauge is green, load recipe
7. Start recipe
8. Wait until you can press the hold button
9. Wait for the PC to fully pump down
10. deselect hold button to start deposition steps
11. Monitor current/power on the source crucible
12. move the beam around to reduce the power/current → Set pos button in Ether Cat device
13. after deposition steps are finished, let the PC vent.
14. Unload sample → unlatch pullout dome and take out Sample
15. Take off carrier wafer
16. After closing the door, go to automap, press Pump PC, select "Down open" and start

Remarks on recipes

17. Turn log off

- Make sure to always include a pump down step and a vent step wrapping your deposition step(s). This allows you to take advantage of the **hold** button to have control over your process.
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